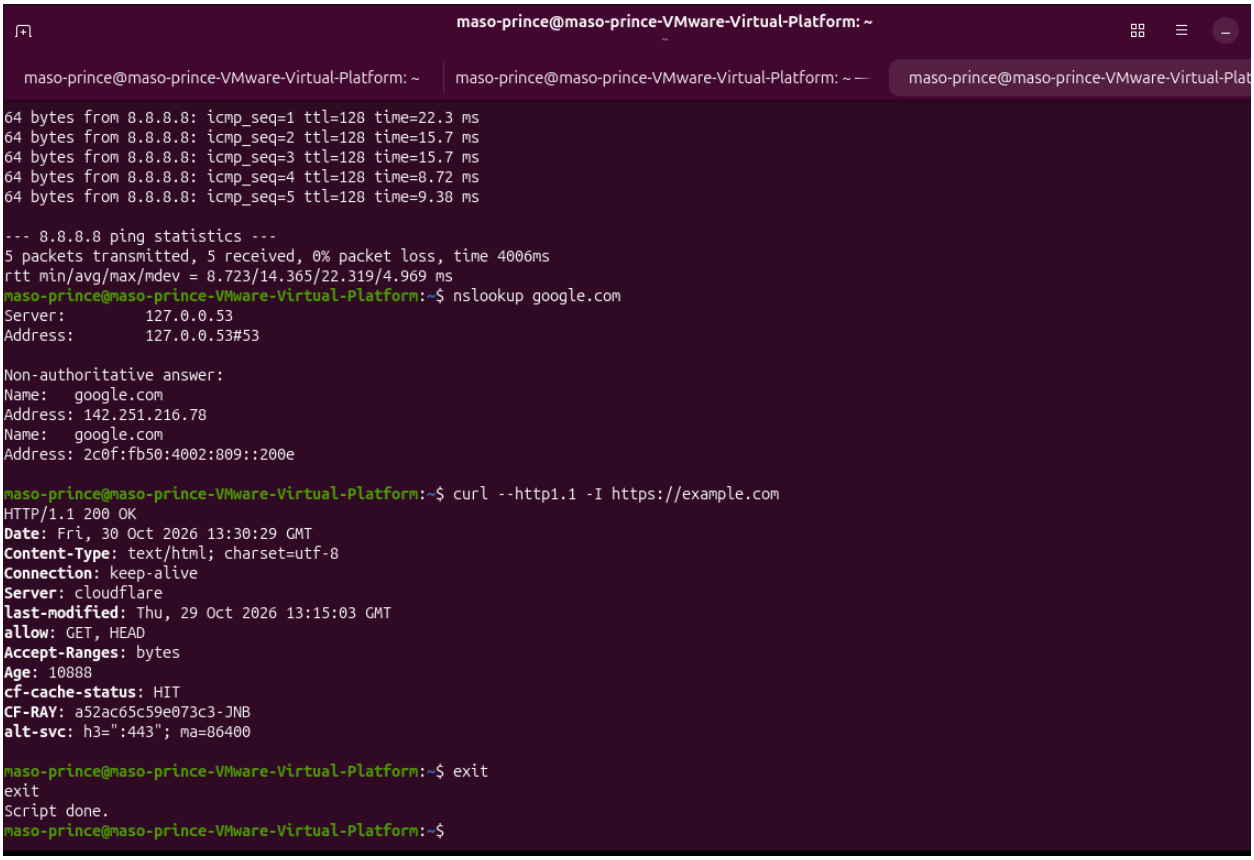
NET732

Week 2: Packet Analysis

Practical

1. Preparing Wireshark



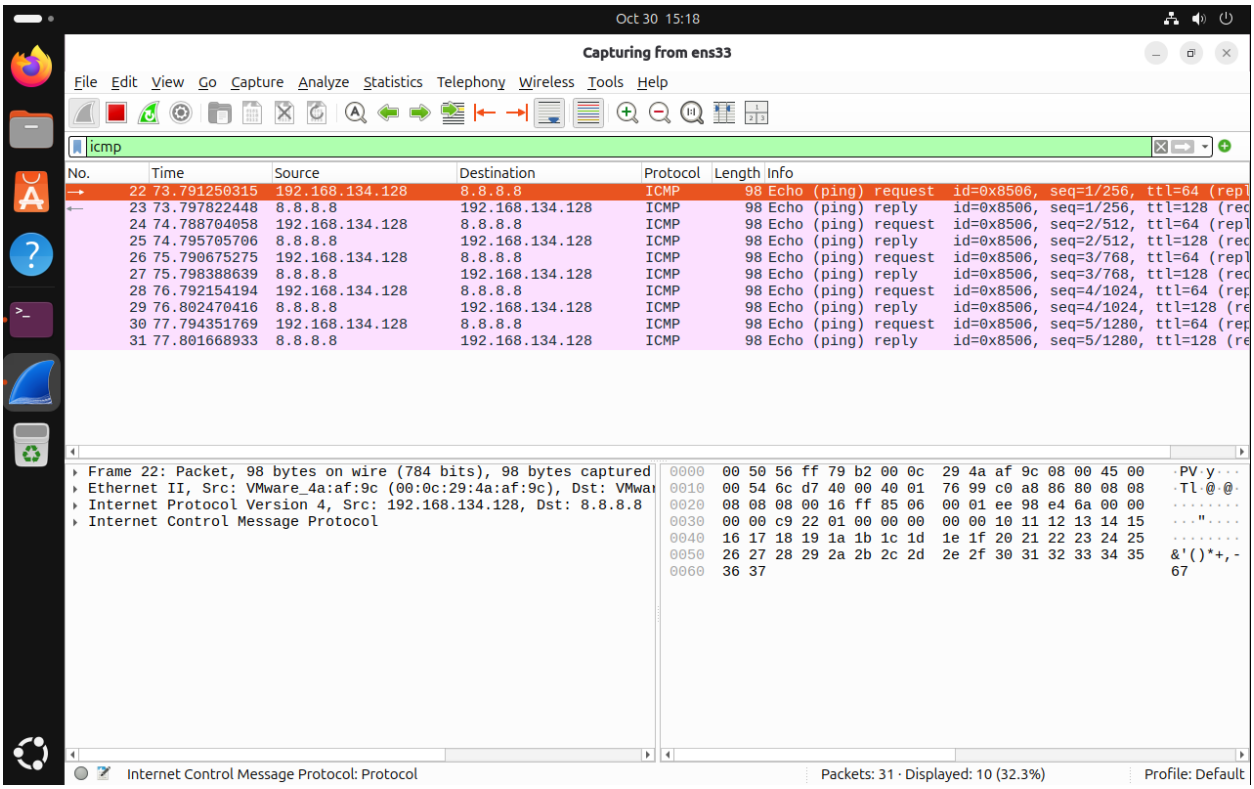
I opened Ubuntu and launched Wireshark. I selected my active network interface to start capturing network traffic.

2. Capturing ICMP traffic

I used the following command to generate ICMP traffic:

ping -4 -c 5 8.8.8.8

In Wireshark, I applied the icmp filter to view the packets.



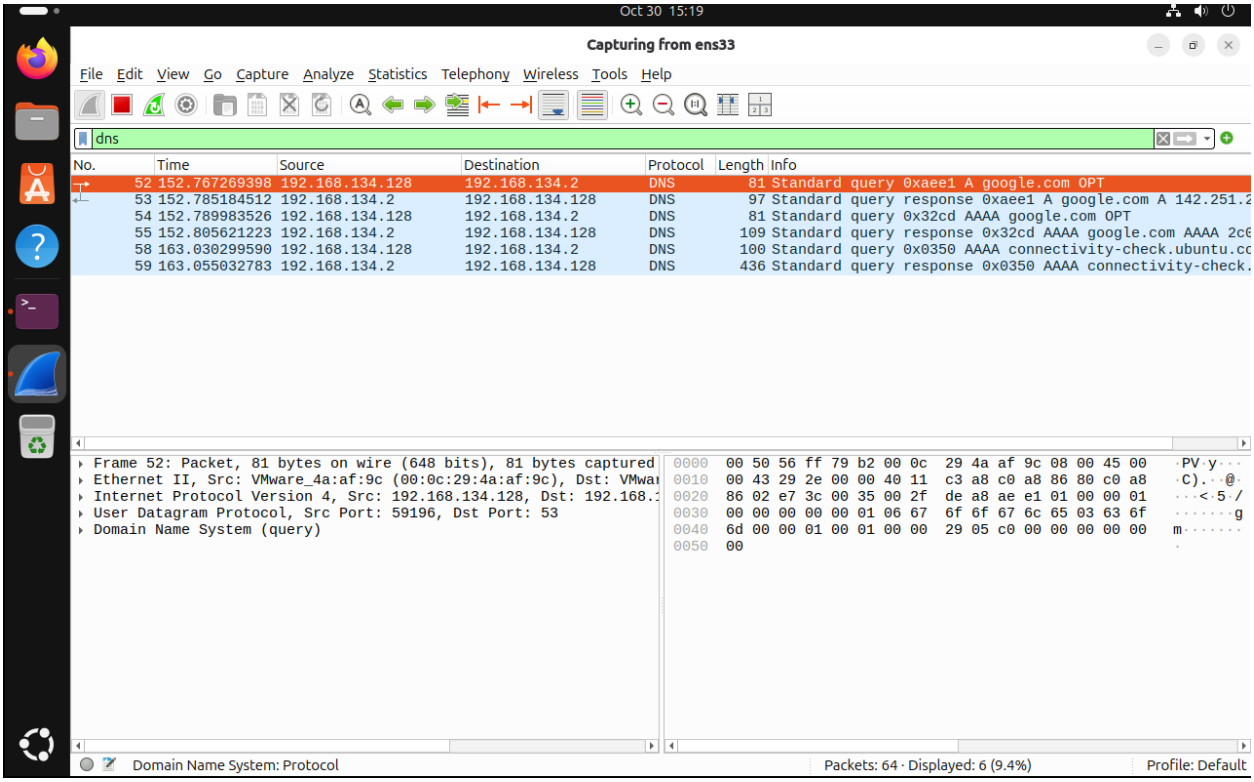
I observed Echo Requests and Echo Replies, which are used to test network connectivity.

3. Capturing DNS traffic

I used the following command:

nslookup google.com

I applied the dns filter in Wireshark to view DNS queries and responses.



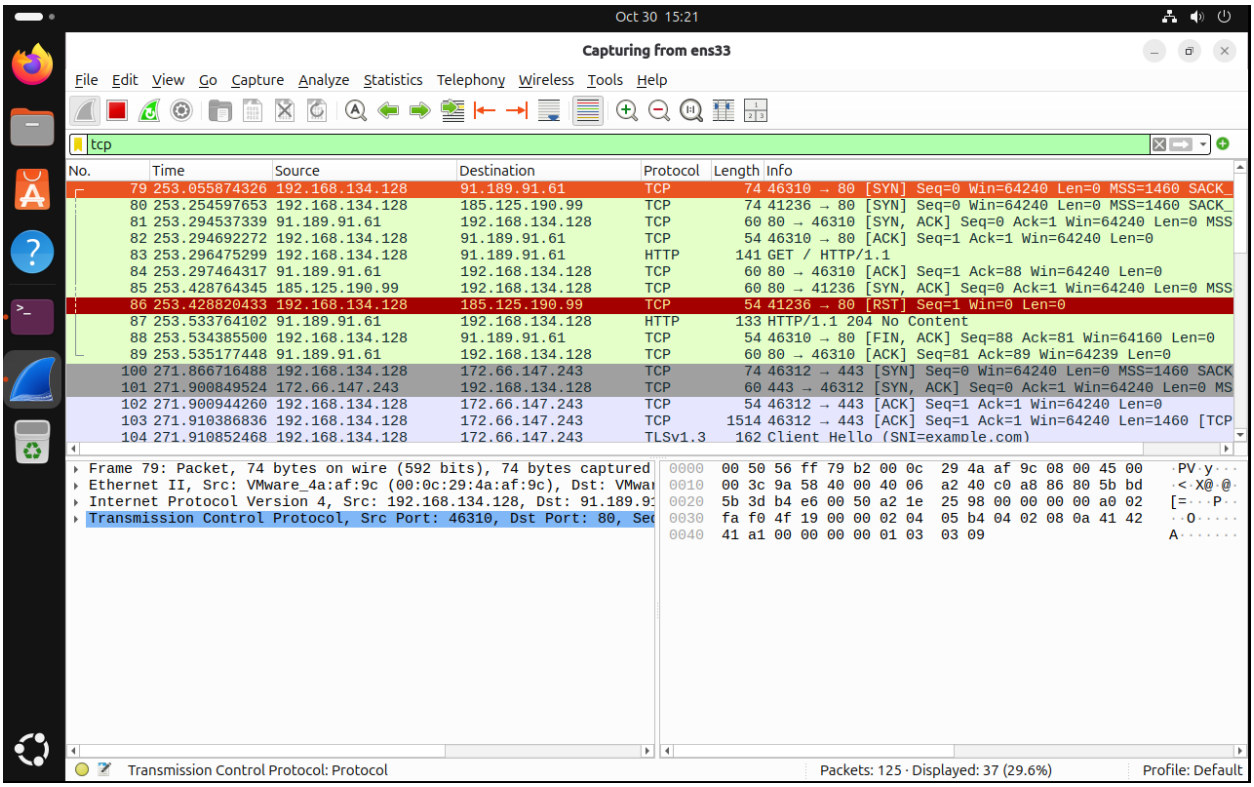
I observed how DNS translates domain names into IP addresses.

4. Capturing TCP traffic

I used the following command:

curl --http1.1 -I https://example.com

I applied the tcp filter in Wireshark to view TCP packets.



I observed the TCP three-way handshake:

- SYN – Requests a connection.

- SYN-ACK – Acknowledges the request.

- ACK – Establishes the connection.

5. Saving packet captures and logs

I saved my Wireshark capture as:

CyberOps_Week2_Capture.pcapng

I also recorded my terminal commands and outputs using:

script -a ~/CyberOps/Week2/commands.log

```
maso-prince@maso-prince-VMware-Virtual-Platform:~$ mkdir -p ~/CyberOps/Week2
maso-prince@maso-prince-VMware-Virtual-Platform:~$ script -a ~/CyberOps/Week2/commands.log
Script started, output log file is '/home/maso-prince/CyberOps/Week2/commands.log'.
maso-prince@maso-prince-VMware-Virtual-Platform:~$ date
Fri Oct 30 03:29:05 PM SAST 2026
maso-prince@maso-prince-VMware-Virtual-Platform:~$ hostname
maso-prince-VMware-Virtual-Platform
maso-prince@maso-prince-VMware-Virtual-Platform:~$ ping -4 -c 5 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=22.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=15.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=15.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=8.72 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=128 time=9.38 ms

--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 8.723/14.365/22.319/4.969 ms
maso-prince@maso-prince-VMware-Virtual-Platform:~$ nslookup google.com
```

6. Analysis and conclusion

During this practical, I captured and analysed ICMP, DNS and TCP traffic using Wireshark. I learned how these protocols are used for network communication.

This relates to SOC operations because security analysts monitor network traffic to identify suspicious activities, investigate communication and detect possible security threats.

I successfully completed the packet analysis activities and collected screenshots, command outputs and packet captures as evidence. My next step is to improve my ability to identify unusual network traffic.